Dynamically Morphing NACA 2416 Aerofoil to Optimize Lift to Drag Ratio

Problem Statement

Aircraft rely on aerofoils to generate lift, but aerofoils maintain a fixed shape during flight. A single shape can only be optimized for a limited set of conditions. Because aircraft operate across many different flight conditions, this leads to reduced efficiency. Pressure and temperature changes aerofoil performance across different regions. International and long distance flights face a multitude of flight conditions they are not optimized for.

Engineering Goal

Design a morphing aerofoil that can dynamically adjust its camber and chord across a wide range of airspeeds and angle of attacks and fabricate the foil with structural integrity and flexibility to sustain controlled deformations.

Materials

PLA was used for the rigid internal structure of the aerofoil.
TPU 95A was used for flexible control surfaces to allow the foil to morph.
High torque servo motors connected with string were used to drive the morphing aerofoil.
An Arduino Uno, breadboard, and 6V power supply were used to control the system and map the most efficient morph states depending on entered conditions.
All components were fabricated using multi material 3D printing on a Prusa 5T XL Input Shaper.

Existing Research

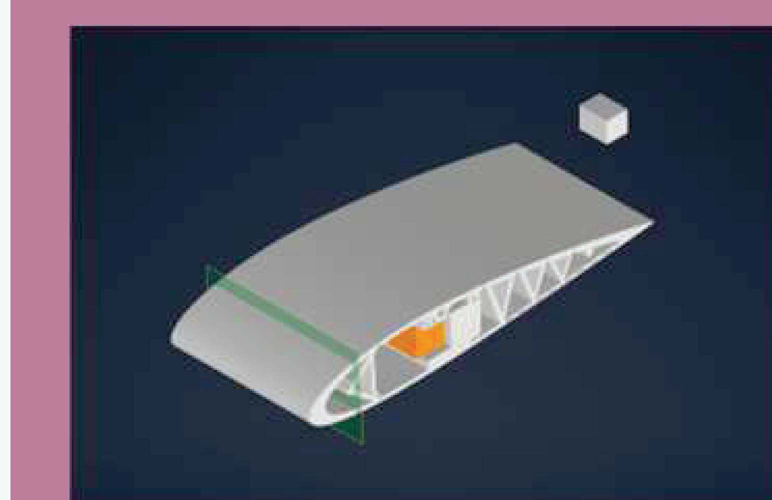
Research shows that changing aerofoil geometry can improve lift to drag ratio across different flight conditions. Aerodynamic principles show that camber and angle of attack directly affect lift generation and drag forces. Wind tunnel testing requires proper scaling to maintain Reynolds number similarity.

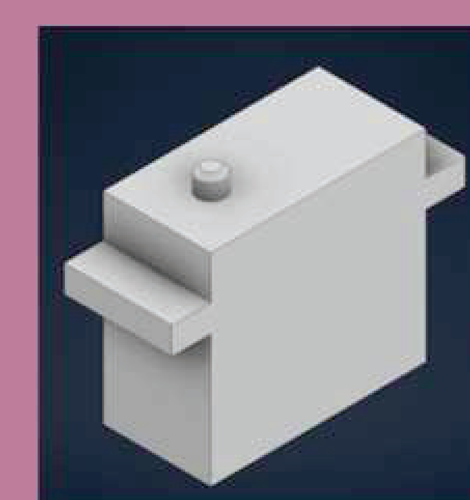
NASA developed the Adaptive Compliant Trailing Edge (ACTE) aerofoil to deform the wing and reduce drag caused by flaps. Research by Z. A. Rana and colleagues showed morphing trailing edge aerofoils can have positive impacts on aerodynamic efficiency.

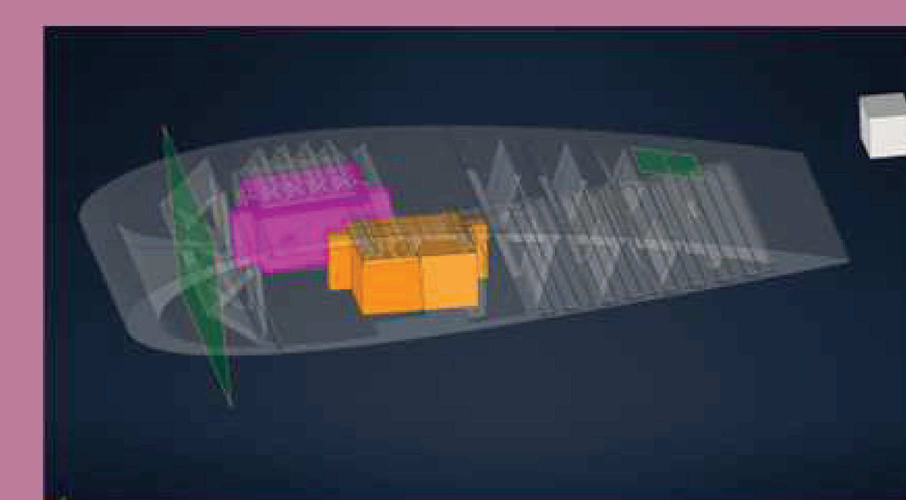
Design Process

The design began with a standard NACA 2416 aerofoil as a baseline geometric structure. A 3D CAD model was developed in Autodesk Inventor Professional 2026. A leading edge actuator and flexible trailing edge were integrated. Servo motors were used to control both control surfaces. Stress analysis was conducted to validate deformation and structural integrity using Von Mises simulations. Design iterations were made to improve morphing behavior and force distribution. The leading edge and trailing edge flex surfaces were directly integrated into the frame with a continuous seal around the foil for smooth aerodynamic deviations.

Cad Models


Morphing Aerofoil

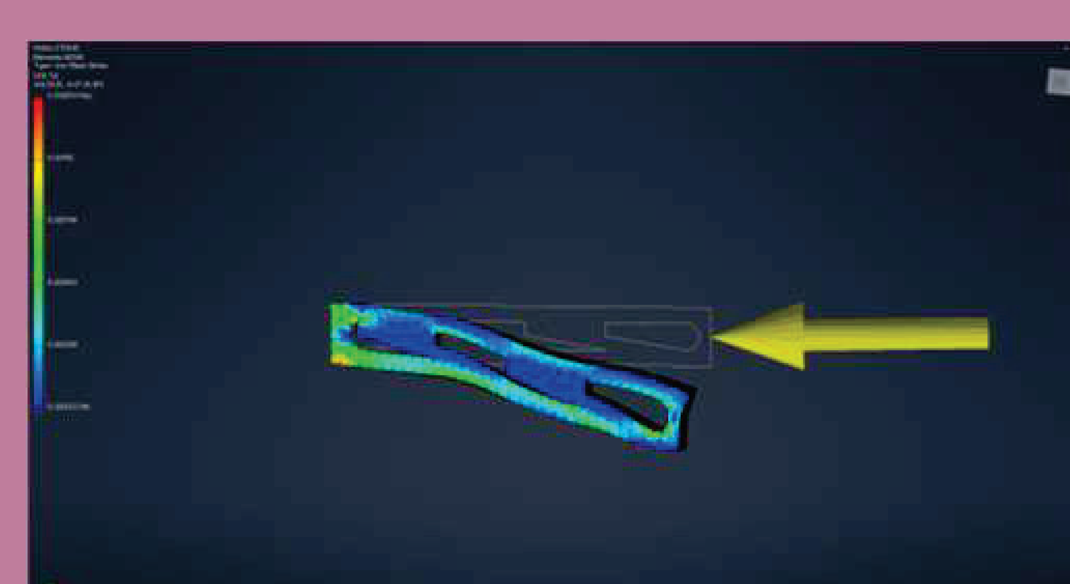

Servo

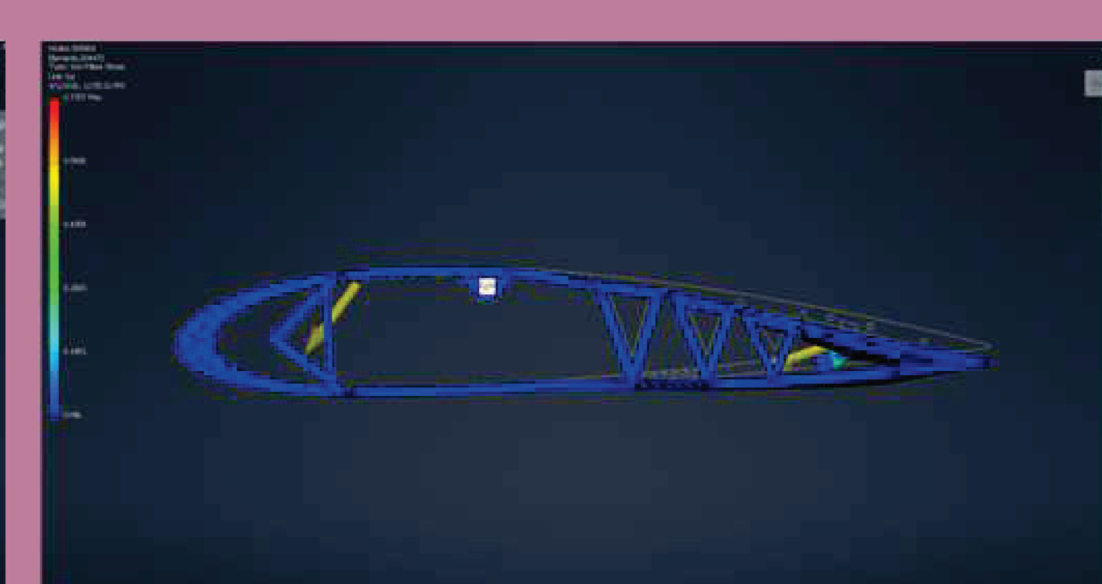

Morphing Aerofoil + Servos

Modifications

The trailing edge was redesigned using a compliant structure to allow smoother deformation. The leading edge actuation system was adjusted to improve force application. An internal truss system was added to better distribute force loads. The internal structure of the material was changed from a grid to a gyroid structure to equally distribute force in a spherical gradient to prevent interference with the flex action. Servo housings were refined to ensure proper fit and stability. The servo housing has 0.1 mm of tolerance for a press fit. The truss structure inside the trailing edge was modified to remove connection on the bottom surface to allow for a more flexible guided joint.

Cad Models


Compliant Structure Von Mises Test


Full Aerofoil Von Mises Test

Results

A static NACA 2416 aerofoil was successfully fabricated. The morphing aerofoil design reached a full scale Mk. 3 1 cm model. 3D printing produced high fidelity surfaces with minimal drag impact. Preliminary models were used to validate scaling and calibration. The Mk. 3 1 cm model exhibited a high level of controlled flexing in the leading and trailing edge.

Conclusions

The project successfully demonstrated that a morphing aerofoil system is mechanically feasible with the current design. A full model would allow Both simulation and physical models showed controlled shape change. However, aerodynamic performance improvements were not quantified due to incomplete testing.

Future Research

Future development will focus on completing a full-scale morphing aerofoil. Wind tunnel testing will be used to collect lift and drag data. Control systems will be improved to better map flight conditions to morphing states. This testing will allow the direct comparison between morphing and static NACA2416 aerofoil performance. Additionally, computational fluid dynamics like ANSYS Fluent and Autodesk CFD should be used to estimate the flight characteristics to refine the morphing mechanics to be more effective. Further work could apply this system to aircraft, drones, and wind turbines.

Aerodynamics. (2018). *Funk & Wagnalls New World Encyclopedia*, 1.

Rana, Z. A., Mauret, F., Sanchez-Gil, J. M., Zeng, K., Hou, Z., Dayyani, I., & Könözsy, L. (2022). Computational analysis and design of an aerofoil with morphing tail for improved aerodynamic performance in transonic regime. Aeronautical Journal, 126(1301), 1144-1167. https://doi.org/10.1017/aer.2021.122

NASA. (n.d.). Wind tunnel fundamentals. Stack Exchange. (n.d.). Scaling for wind tunnel results.